

ZHARYK, KazakhstanWMO: **354970**Lat: **48.850N**Lon: **72.867E**Elev: **2152**StdP: **13.59**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-15.7	-27.4	1.3	-21.1	-21.5	1.8	-14.9	26.4	13.1	20.5	19.2	1.4	70	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	89.4	63.2	86.1	61.9	83.0	60.9	65.8	82.3	64.3	80.7	62.9	78.7	4.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.0	86.7	70.9	58.7	79.8	69.3	56.6	74.0	69.0	31.9	83.0	30.6	80.8	29.5	79.0	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	17.4	13.9	11.7	DB	-27.8	94.3	6.5	3.9	-32.4	97.1	-36.2	99.4	-39.8	101.5	-44.5	104.3
(5)				WB	-28.0	69.4	6.4	5.0	-32.6	72.9	-36.3	75.8	-39.9	78.6	-44.6	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	8.3	10.3	24.2	44.4	56.1	65.9	67.7	65.6	54.7	40.9	25.1	12.9
	DBStd	23.88	11.20	11.56	11.71	10.38	7.99	7.34	6.68	7.56	9.12	9.89	12.21	11.75
	HDD50	5724	1292	1110	799	223	34	1	0	2	55	312	746	1150
	HDD65	9556	1757	1530	1263	618	288	79	48	86	328	748	1196	1615
	CDD50	2015	0	0	1	56	225	478	548	485	194	28	0	0
	CDD65	372	0	0	0	1	13	105	131	104	18	0	0	0
	CDH74	4358	0	0	0	28	208	1177	1425	1242	275	3	0	0
	CDH80	1399	0	0	0	4	34	352	506	429	74	0	0	0
Wind	WSAvg	4.2	4.5	4.9	5.2	4.5	4.5	3.8	3.6	3.5	3.4	3.7	3.8	4.6
Precipitation	PrecAvg	13.6	1.1	0.9	1.1	0.9	1.3	1.3	1.9	0.7	0.6	1.1	1.3	1.3
	PrecMax	18.4	2.3	1.7	2.6	2.4	3.5	4.4	5.5	1.6	2.2	2.8	2.8	4.0
	PrecMin	9.2	0.2	0.3	0.2	0.0	0.4	0.0	0.5	0.0	0.0	0.2	0.4	0.4
	PrecStd	2.6	0.6	0.4	0.6	0.6	0.9	0.9	1.1	0.5	0.5	0.7	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	36.0	57.2	77.4	84.3	91.1	94.6	92.8	87.2	72.8	55.8	35.9
		MCWB	31.9	32.4	44.6	54.6	60.9	65.5	64.9	63.0	59.2	51.7	45.2	32.5
	2%	DB	30.6	32.5	48.4	71.0	79.2	87.3	89.6	88.6	81.2	67.3	49.6	33.2
		MCWB	28.4	29.3	41.0	52.1	58.1	62.8	63.9	62.2	56.5	48.6	42.4	30.4
	5%	DB	26.9	29.3	42.8	66.3	75.7	84.1	85.4	84.8	76.3	61.4	44.3	30.6
		MCWB	24.7	26.6	37.8	49.8	56.1	61.4	62.7	61.2	55.2	45.9	38.7	28.1
	10%	DB	24.0	26.5	38.9	61.6	72.3	81.0	81.8	81.1	71.6	56.0	39.9	27.8
		MCWB	21.9	24.1	35.1	47.7	54.6	60.7	61.7	60.2	53.1	44.0	35.1	25.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	33.2	45.6	56.3	64.1	73.4	69.5	66.3	61.9	53.4	48.0	33.9
		MCDB	34.1	35.3	54.3	73.1	79.0	79.6	81.5	83.8	81.5	68.1	52.1	35.4
	2%	WB	28.5	29.7	41.4	53.4	59.9	65.5	66.1	64.2	58.5	50.0	42.8	30.8
		MCDB	30.8	31.9	47.3	69.0	74.3	80.9	83.0	83.2	77.6	64.1	49.3	32.5
	5%	WB	25.0	26.8	38.1	50.9	57.7	63.6	64.6	62.8	56.0	47.4	39.1	28.3
		MCDB	26.8	29.0	42.6	64.2	71.8	79.4	80.4	80.8	73.3	58.6	44.0	30.6
	10%	WB	22.1	24.1	35.2	48.5	55.8	61.9	63.3	61.3	53.9	45.0	35.5	25.3
		MCDB	24.0	26.4	38.6	59.8	69.6	77.5	78.1	78.3	69.6	54.4	39.7	27.5
Mean Daily Temperature Range	5% DB	MDBR	15.6	17.0	15.4	20.7	23.1	22.8	21.8	23.6	23.7	19.2	14.6	14.6
		MCDBR	12.1	14.8	17.2	27.1	27.6	26.3	26.6	27.8	29.1	27.4	18.9	14.0
		MCWBR	11.1	13.3	12.1	14.2	12.9	11.0	10.1	10.3	12.7	14.4	13.1	11.9
	5% WB	MCDBR	11.7	13.8	16.1	25.3	24.1	23.4	23.4	25.3	27.4	23.9	17.3	13.1
		MCWBR	11.0	13.1	11.9	13.8	12.6	11.8	10.8	10.9	13.0	13.5	13.0	11.9
Clear-Sky Solar Irradiance	taub	0.285	0.317	0.367	0.401	0.405	0.405	0.424	0.386	0.341	0.332	0.318	0.289	
	taud	2.090	2.026	2.016	2.139	2.187	2.254	2.208	2.295	2.378	2.323	2.175	2.094	
	Ebn at Noon	236	255	262	267	271	272	264	270	272	252	221	215	
	Edh at Noon	27	37	45	44	44	42	43	38	31	28	25	24	
All-Sky Solar Radiation	RadAvg	538	920	1313	1604	2023	2129	1965	1823	1410	803	436	390	
	RadStd	60	40	101	137	158	145	144	124	92	97	39	43	

Historical Trends

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